
a
SUM add
1
7
8
g = 6, carry 1
one token
1 + 7 + 8 = 16
242 views
MUL multiply
7
6
4
6 4 A
digit carry next
7 × 6 + 4 = 46
1,553 views in five steps
SUB subtract
1
3
8
4 1 S
digit borrow next
3 − 8 − 1 = −6
242 views
EST estimate a quotient digit
4
7
6
7 – M
guess next
⌊47/6⌋ = 7
400 views
DEC lower the guess
7
6 – C
digit next
7 − 1 = 6
9 views
b
4,100 ÷ 59
quotient
dividend
carry
divisor
carry
scale
guess × divisor
borrow
remainder
8
4 1 0 0
5 9
4
1
4 3 8
the borrow out of the top column says the guess 8 is too high
Algorithm D: every forward pass of the run, left to right
EST
MUL
SUB
DEC
SUM
0
8
7
6
9
guess 0, 8
8 × 59 from 410
too high: 7
add back
too high: 6
add back
guess 9
9 × 59 from 560
square root of 2,000,000 = 1,414: 56 forward passes
EST
MUL
SUB
DEC
ADD
c
four jobs at once: 32 forward passes, against 73 one after another
8051 ÷ 97
√ 52441
4813 × 276
90210 − 31337
11 alone
32 alone
24 alone
6 alone
forward pass
0.065 s a pass
+ 0.037 s a view
median ratio 1.01
measured on the M4 (s)
cost model (s)
0
6
12
18
d
50/50
50/50
50/50
50/50
12 s
seconds
÷ to 10 places
√ to 6 places
gcd
signed ±